

Thermo Scientific™ HyPerforma™ 5:1 S.U.B. 500 L - 100-500 L working volume

thermo-sub-500

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	100 L	Source-supported
Maximum working volume	500 L	Source-supported
Nominal vessel / bag size	500 L	Source-supported
Total vessel volume loaded into the form	660 L	Source-supported
Initial working volume loaded into the form	100 L	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	1.5	Source-supported
Internal / chamber diameter	0.7556 m	Source-supported
Liquid height at maximum working volume	1.13 m	Source-supported
Impeller / mixing type	Pitched-blade turbine	Source-supported
Impeller count	1	Source-supported
Impeller diameter	0.251 m	Source-supported
Power number	2.1	Source-supported
Initial agitation	59 rpm	App assumption
Maximum agitation	59 rpm	App assumption
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	No	Source-supported
Maximum pressure	0.03 bar(g)	Source-supported
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 100 L

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 100 L.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

[Manufacturer publication](#)

Maximum working volume - 500 L

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

[Manufacturer publication](#)

Nominal vessel / bag size - 500 L

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

[Manufacturer publication](#)

Total vessel volume loaded into the form - 660 L

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 660 L.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

[Manufacturer publication](#)

Initial working volume loaded into the form - 100 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: $\min(100 \text{ L}, 500 \text{ L}) = 100 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 1.5

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.5.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

Manufacturer publication

Internal / chamber diameter - 0.7556 m

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.7556 m.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

Manufacturer publication

Liquid height at maximum working volume - 1.13 m

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.13 m.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

Manufacturer publication

Impeller / mixing type - Pitched-blade turbine

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

Manufacturer publication

Impeller count - 1

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

[Manufacturer publication](#)

Impeller diameter - 0.251 m

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.251 m.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

[Manufacturer publication](#)

Power number - 2.1

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.1.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

[Manufacturer publication](#)

Initial agitation - 59 rpm

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 59 rpm.

Limitations: Starting RPM lowered to the selected constant low-fill ceiling; not a separately sourced process setpoint.

Maximum agitation - 59 rpm

App assumption | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 119; Table 3.3, 500 L, 20% volume / 20 W/m³

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 59 rpm.

Limitations: The app has one constant ceiling, but PDF p. 119 limits 20-50% fill to 20 W/m³. Select the published 20%-fill nominal RPM across the range, rather than misrepresenting the motor maximum as valid at every fill. Full fill allows higher RPM. Published RPM values are rounded and assume water-like fluid.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 119](#)

[Manufacturer publication](#)

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - No

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151; Table 4.7, 500 L, printed p. 146

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: No.

Limitations: DOC0022 Rev. H (February 2021). Direct BPC chamber diameter and rated-volume liquid height. One three-blade 45-degree down-pumping impeller. Np 2.1 is manufacturer-calculated, not an independent measured power coefficient. Absolute dimensions take precedence over rounded ratios; 1.5 is the rounded fluid H/D.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 151](#)

Manufacturer publication

Maximum pressure - 0.03 bar(g)

Source-supported | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 9; BPC operating limits, printed p. 4

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.03 bar(g).

Limitations: Explicit 0.03 bar (0.5 psi) BPC pressure ceiling. Bar value retained; not water-jacket or sparger-inlet pressure.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 9](#)

Manufacturer publication

Default operating pressure - 0 bar(g)

App assumption | Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 9; BPC operating limits

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0 bar(g).

Limitations: Atmospheric app starting value; replaces a default above the published BPC limit. Not a manufacturer culture setpoint.

Local original: [Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide - PDF page 9](#)

Manufacturer publication

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: `maxOxygenFraction = 100`

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

These presets use a generic mammalian-cell-culture profile.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Detailed guide motor speed ceiling: 150 rpm. Simulator constant ceiling: 59 rpm, selected for minimum fill. The actual operating envelope needs a fill-dependent governor; no such governor was added. Table 3.3 p. 119 uses 39.8 cm for the 2000 L impeller, versus 39.7 cm in Table 4.11; the detailed hardware dimension is selected. Product-contact film remains configuration-dependent.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Thermo HyPerforma 5:1 Single-Use Bioreactor User Guide

thermo-sub-5to1-user-guide-b628fb94036b.pdf

SHA-256: b628fb94036bb4cc841115976bc634f3234d7d1ab87f8d41796bb189d89897bd